

CSE212: SOFTWARE DEVELOPMENT METHODOLOGIES

YEDITEPE UNIVERSITY

SPRING 2025

TERM PROJECT – DUE DATE JUNE 7TH, 2026

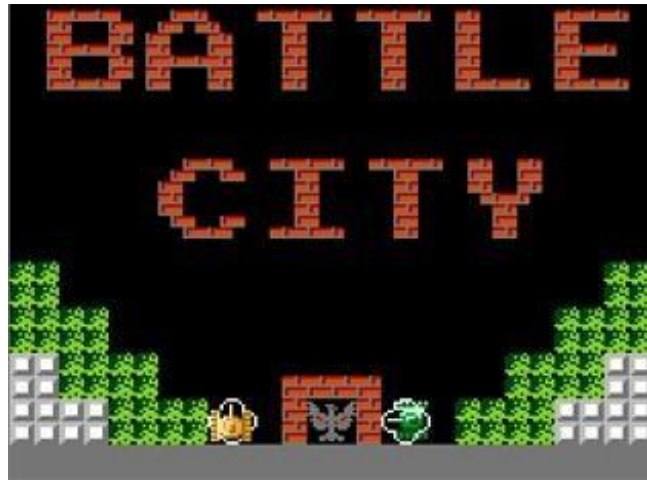
As a term project, you are required to develop a *Battle City* (Tank 1990) game with multiple difficulty levels. The game should be simple and easy to use with the help of WASD on the keyboard – up/down (W and S keys) and left/right(A and D keys). Your application should be able to keep scores of users and store them on a separate CSV file.

You can use the following statements and figures as guidelines:

- Your application should have a menu on the window frame, which contains *New Game*, *Map Editor*, *Options*, *High Scores*, *Help*, *About* and *Exit*.
- For usability purposes, you are required to implement a graphical user interface (GUI) for your application, meaning that the game should have its own window frame.



(a)



(b)

Fig. 1: Example Battle City game screenshot

- At the end of each game, it should ask for the user's name and record his/her details (*name*, *time*, *date* and *score*) to a file.

- At any time, users should be able to pause the game by clicking on the appropriate button on the GUI, such as the Pause button.

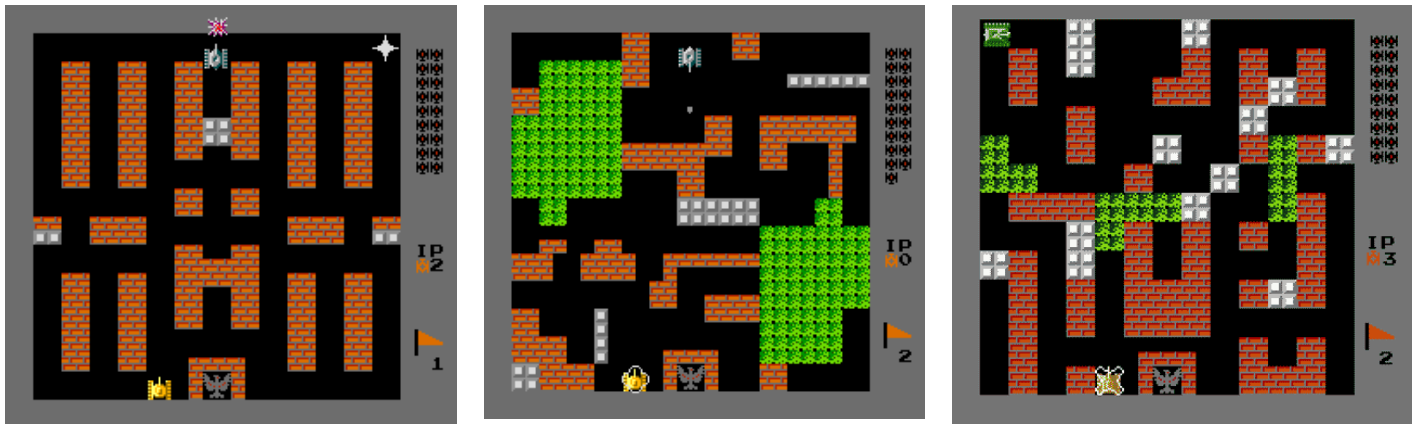


Fig. 2: Three levels with different designs

- The game should have at least *three* (see Image Pack) levels of difficulty (see Fig. 2). You can increase the difficulty by changing the shape of the terrain, the type of bricks used and increasing the speed of the enemy tanks.
- Your application should show the number of lives you have on the right hand panel in the top right corner of the game window (Fig 2).
- The player, controlling a tank, should try to destroy enemy tanks in each level, which enter the playfield from the top of the screen.
- The enemy tanks should attempt to destroy the player's base (represented on the map as a bird, eagle or Phoenix), which is positioned at the bottom centre of the screen surrounded with bricks, as well as the human-controlled tank.
- A level is completed when the player destroys all 20 enemy tanks, but the game ends if the player's base is destroyed (the bird icon, for instance, in Fig. 2) or the player loses all available lives (shown on the right hand of the screen).
- In the game, the brick walls can be destroyed by having either the player's tank or an enemy tank shoot at them, steel walls can only be destroyed by the player if he/she has collected three or more power-up stars.
- The game should end when either the player's tank is shot or the base (eagle) (bird icon in Fig. 2) is destroyed by enemy tanks.

- The tanks should be able to move under the green bushes in the terrain and these bushes should hide the player's tank and enemy tanks to add an extra level of difficulty.
- You should only use concepts discussed within the course scope, such as Polymorphism, Collections, Inheritance, and Threads. Any topics not included in the course work will yield to additional questions during the demonstration phase and failing to answer them will result in penalties in the grading.
- You should implement the following power ups and bonuses for your game:
 - ⤴ **Tank** symbol gives an extra life;
 - ⤴ **Star** improves player's Tank (having one star make shots faster, having two stars allow two simultaneous shots, having three stars allow the player to destroy steel);
 - ⤴ **Bomb** destroys all visible enemy tanks;
 - ⤴ **Clock** freezes all enemy tanks for a period of time,
 - ⤴ **Shovel** adds steel walls around the base for a period of time and;
 - ⤴ **Shield** makes the player's tank invulnerable to attack for a period of time.

You should also have a map editor (*Map Editor*) functionality so that the player could create their own maps with different difficulties using various elements, such as green bushes, water, brick walls, and steel walls.

- **[Bonus]** Your application could also have the audio effects during the game and any bonus item collected by the tank.
- Users should also be able to see the high score list, which contains the list of ten highest scores by clicking the appropriate menu item (*High Score*).
- The help menu option should provide some help on how to play the game (*Help*) and the About (*About*) option should pop-up a small window that contains information (*Name, Surname, School Number and Email*) about the application developer.

- You can access the image pack for your game at the following address:
<https://yulearn.yeditepe.edu.tr/mod/resource/view.php?id=343464>
- Also you can watch the game play and create sound bites for your game using the following video
<https://www.youtube.com/watch?v=Cl79I6ryTYs&t=4211s>

Submit your project source code in a zip file, which has your student number as name, through the YULearn system by the end of Sunday, June 7th, 2026. All submitted source files will be checked for plagiarism - among classmates and with any existing open source code available on the Internet, as well as any generated AI code. Furthermore, all students will be required to demonstrate their work for 15 minutes. DO NOT submit somebody else's work.